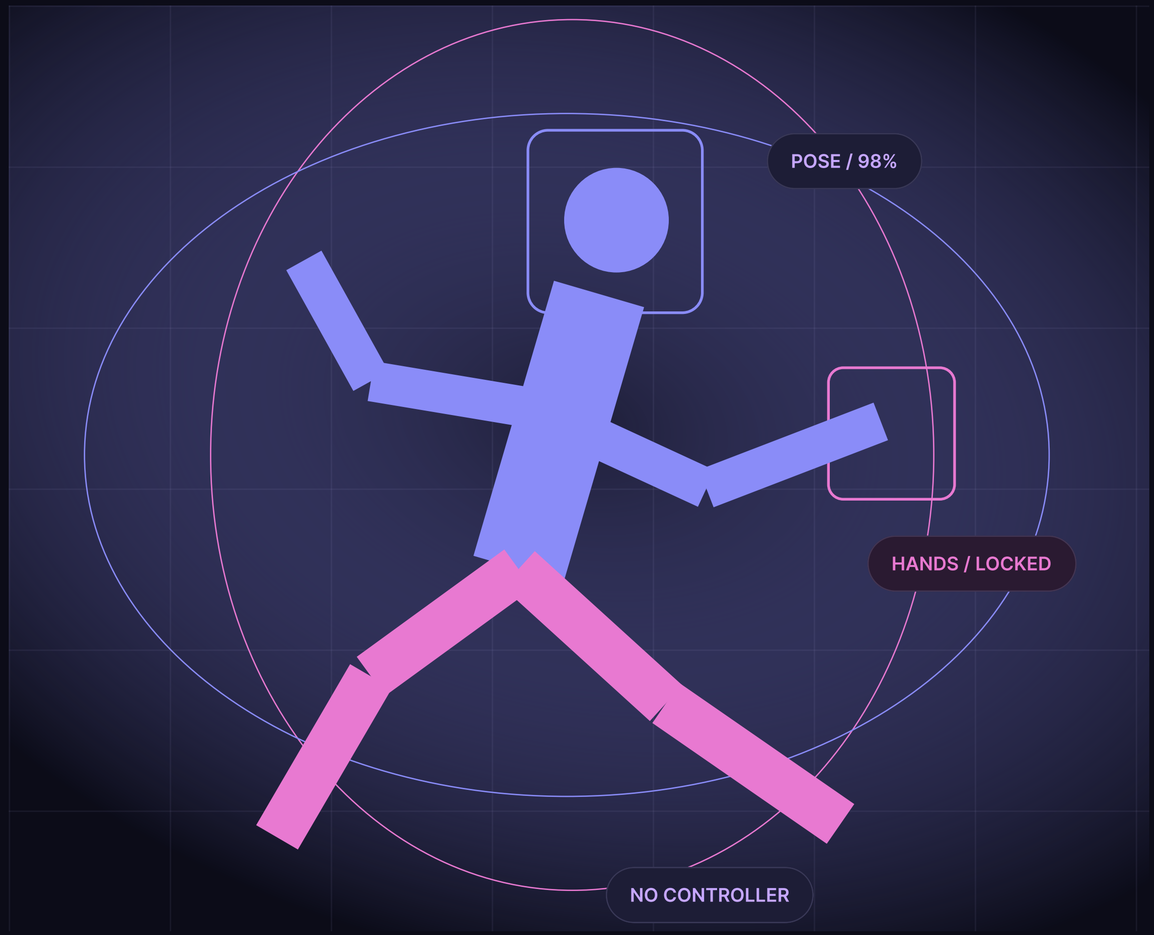
YOUR BODY IS THE CONTROLLER

MOVE. REACT. RUN.

A hands-free endless runner powered by body, hand, and object tracking.

Webcam tracking · Full-body control · Object detection



01

FROM CALIBRATION TO QUIT

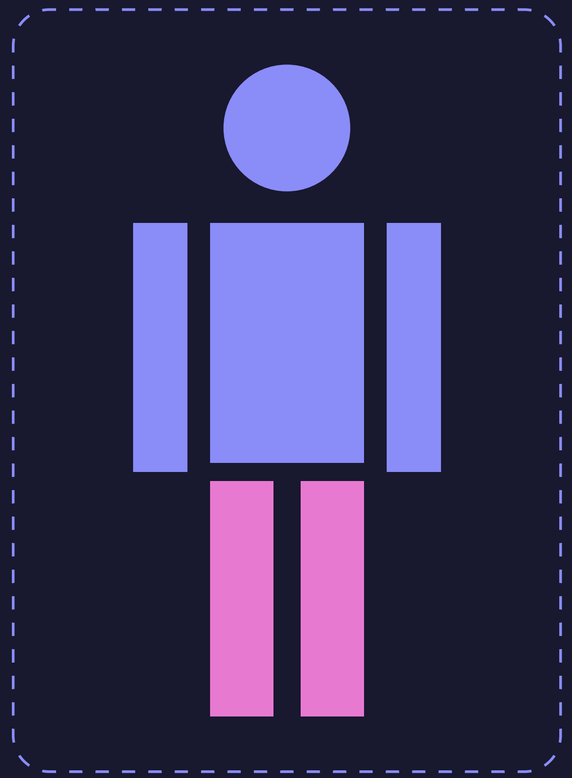
Application Design

Eleven screens. One completely hands-free experience.

AYATLA · HUMAN–COMPUTER INTERACTION

AYATLA

SETUP · 1 OF 1



FULL BODY · SAFE AREA

LET'S GET YOU READY

Step back until your full body is inside the frame.

✓ Camera detected ✓ Body detected ✓ Hands detected

Hold still while the ring fills.

POSE CALIBRATION / AUTOMATIC · NO INPUT REQUIRED

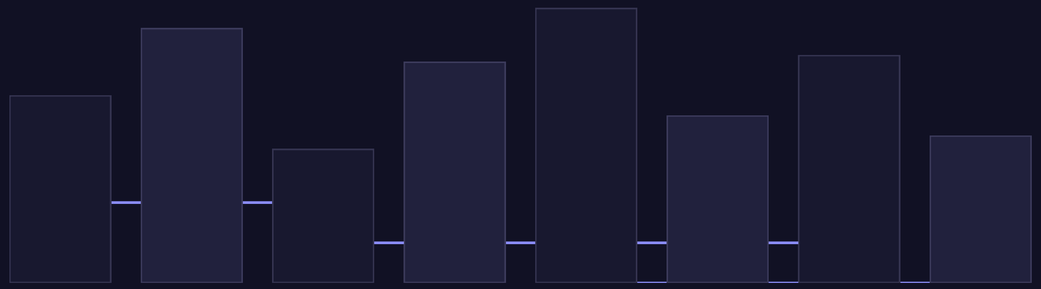
AYATLA

BODY TRACKING ACTIVE

READY TO MOVE?

RUN / THE CITY.

An endless city. A different way to move.



1 SHOW 1 FINGER
Start Game PENDING

2 SHOW 2 FINGERS
Gesture Guide →

3 SHOW 3 FINGERS
Settings →

4 SHOW 4 FINGERS
Quit →

OK to confirm | Hold X to go back

AYATLA

PRACTICE MODE · LIVE

LEARN THE CONTROLS

Your body is the controller.



Completed

Physical jump

→ Character jumps



Completed

Crouch or squat

→ Character slides



Lean left or right

→ Change lane



Open palm

→ Pause or open quick menu



One to four fingers

→ Focus numbered menu option



OK hand sign

→ Confirm selection



Hold an X

→ Go back or cancel



Show tracked object

→ Activate power-up

Try a gesture

Move naturally and hold the pose.

Jump detected

AYATLA

BODY TRACKING ACTIVE

STARTING SESSION

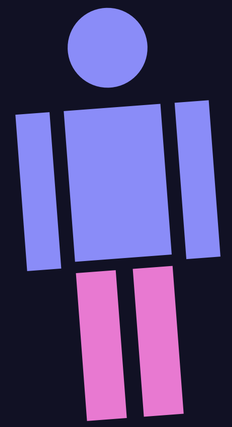
How would you like to play?

PENDING

LEAN LEFT

Sync with Bluetooth

Connect your player profile and keep your scores across sessions.



MOVE LEFT OR RIGHT

LEAN RIGHT

Play as Guest

Start immediately without connecting a player profile.

Score won't be saved

Lean to focus an option

OK sign to confirm

Hold an X to go back

APPLICATION 05 **Gameplay HUD**

1920 × 1080



AYATLA

SESSION PAUSED

GAME PAUSED

Choose with your fingers.

1 SHOW 1 FINGER
Resume PENDING

2 SHOW 2 FINGERS
Restart →

3 SHOW 3 FINGERS
Settings →

4 SHOW 4 FINGERS
Quit →



OK sign

Confirm selection



Hold an X

Go back



Flat palm paused the game

APPLICATION 07 **Game Over**

1920 × 1080

AYATLA

SESSION SUMMARY

RUN COMPLETE

12,840

FINAL SCORE · NEW BEST

BEST	DISTANCE
12,840	2.7 KM

Open palm

→ **Restart**

Finger selection or OK sign

→ **Main menu**

Hold an X

→ **Quit or go back**

AYATLA

BODY TRACKING ACTIVE

PERSONALIZE CONTROL

Settings

Tune the way AYATLA responds to your movement.



Focus a setting with your fingers. Adjust with a deliberate lean.

Subject to change

Gesture sensitivity High

72%

Confirmation hold time 2.0 sec

55%

Sound volume 80%

80%

Camera selection FaceTime HD Camera

Left-handed mode Enabled

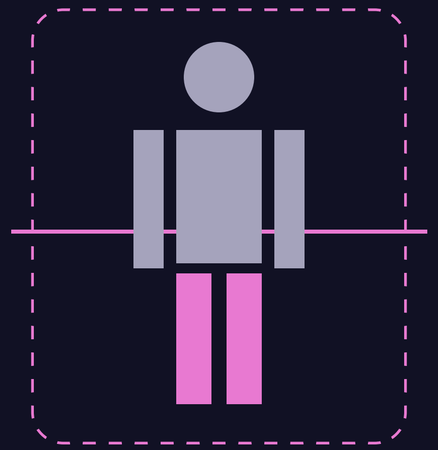
OK sign → Save changes

Open palm → Reset defaults

Hold an X → Go back

AYATLA

TRACKING INTERRUPTED



GAME AUTO-PAUSED

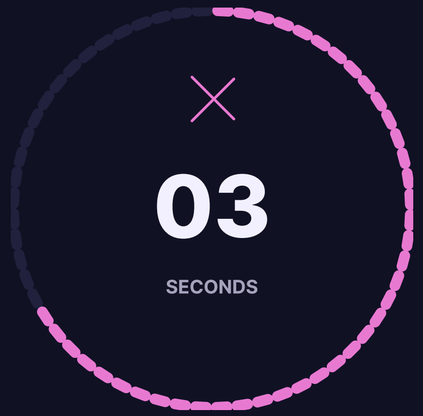
We lost track of you.

Step back into the frame to continue.

Searching for body and hands...

AYATLA

EXIT REQUEST PENDING



CONFIRM QUIT

Hold an X for 3 more seconds to quit.

Lower your arms to cancel and return.

AYATLA

GESTURE READY

HANDS-FREE QUICK MENU

Pause, then choose with your fingers.



HOLD

STEP 01 · DURING GAMEPLAY

Hold one palm flat

Present an open, flat palm to pause the run and reveal the quick menu.

STEP 02 · QUICK SELECTION

1	SHOW 1 FINGER	Resume	PENDING
2	SHOW 2 FINGERS	Restart	→
3	SHOW 3 FINGERS	Settings	→
4	SHOW 4 FINGERS	Quit	→

THEN CONFIRM OR GO BACK



OK sign

Confirm selection



Hold an X

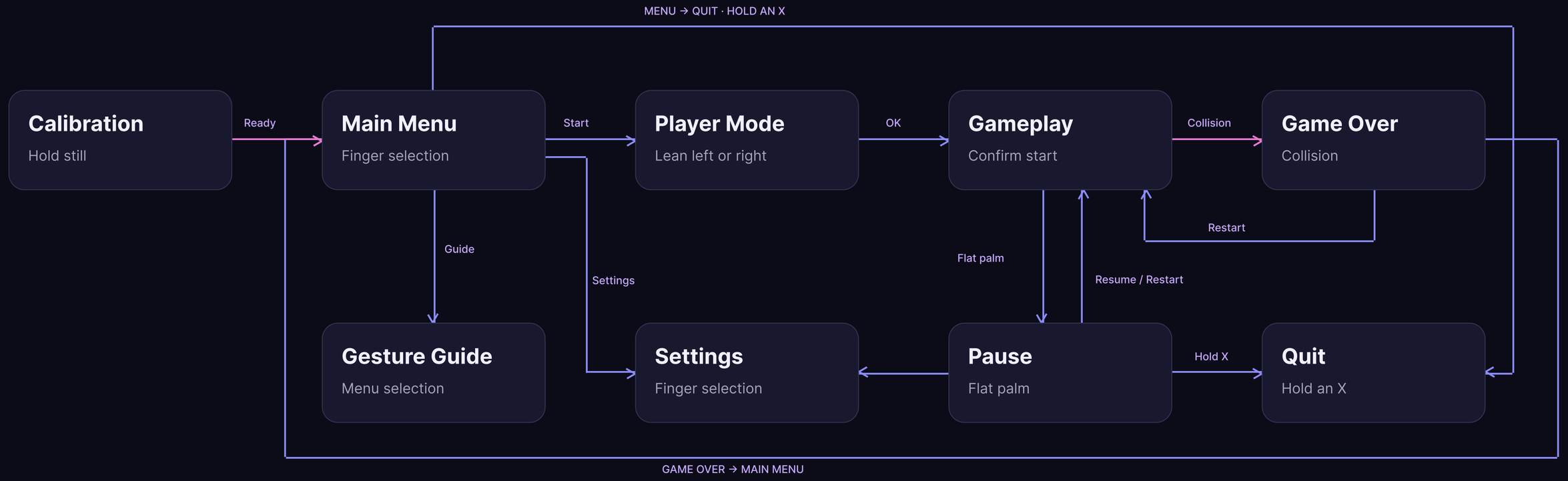
Go back

SYSTEM MAP

Application flow diagram

Gesture transition

System event



Tracking Lost Any screen → auto-pause → previous screen when user returns

02

DESIGNING FOR CONFIDENCE

HCI Design Principles

Making invisible input understandable, predictable, and safe.

AYATLA · HUMAN–COMPUTER INTERACTION

Confidence, by design.

01 Navigation

USED IN
Main Menu · Pause · Quick Menu

→ 1 FOCUS → OK / X BACK

Forward + back controls

WHY
Numbered finger choices, OK confirmation, and the X gesture create a clear route forward and back.

02 Familiarity

USED IN
Gesture Guide · All Menus

PALM / FINGERS / OK

Familiar hand signs

WHY
Recognizable palm, finger-count, OK, and X gestures connect new controls to familiar real-world signs.

03 Consistency

USED IN
Every Screen

OK = CONFIRM · X = BACK

One gesture, one meaning

WHY
The same gesture icons, colors, labels, and meanings are repeated throughout the experience.

04 Error Prevention

USED IN
Pause · Quit · Guest Mode

PENDING → OK TO CONFIRM

Confirm before action

WHY
Pending states, explicit confirmation, a back gesture, and score warnings prevent unintended actions.

05 Feedback

USED IN
Gameplay · Tracking · Selections

JUMP DETECTED +100

Immediate confirmation

WHY
Tracking states, highlighted choices, pending labels, and action toasts show what the system detected.

06 Visual Clarity

USED IN
HUD · Menus · Error Screens

SCORE 12,840 / LIVE

Clear visual hierarchy

WHY
Strong contrast, concise labels, separated content zones, and large gesture cues support quick scanning.

07 Flexibility & Efficiency

USED IN
Mode Select · Pause · Settings

← PROFILE / GUEST →

Multiple efficient paths

WHY
Bluetooth and guest paths support different needs, while finger shortcuts reduce steps for experienced players.

03

UNDERSTANDING THE EXPERIENCE

PACT Analysis & User Scenario

The people, movements, spaces, and technology behind the play.

AYATLA · HUMAN–COMPUTER INTERACTION

PEOPLE / ACTIVITIES / CONTEXT / TECHNOLOGY

Designed around real bodies and spaces.



PACT 01

PEOPLE

- Casual gamers aged 12–35, including students and families.
- Varied mobility, height, handedness, and gaming experience.
- First-time and returning players.



PACT 02

ACTIVITIES

- Short 3–10 minute runs combining play with light exercise.
- Whole-body movement controls gameplay and navigation.
- Pause, restart, settings, and quit without touching a device.



PACT 03

CONTEXT

- Indoor 2 × 2 m clear space under varied lighting.
- Solo play, social viewing, classrooms, or events.



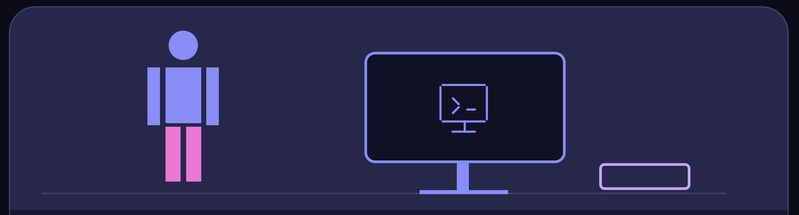
PACT 04

TECHNOLOGY

- 720p or better webcam tracking pose, hands, fingers, and objects.
- Laptop or PC connected to a monitor or TV.
- Optional Bluetooth profile synchronization.
- No keyboard, mouse, touchscreen, or controller input.

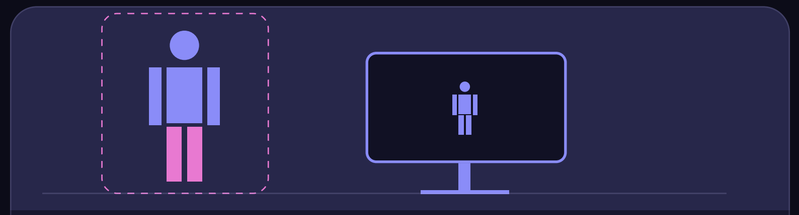
A student’s completely hands-free run

Student
University player



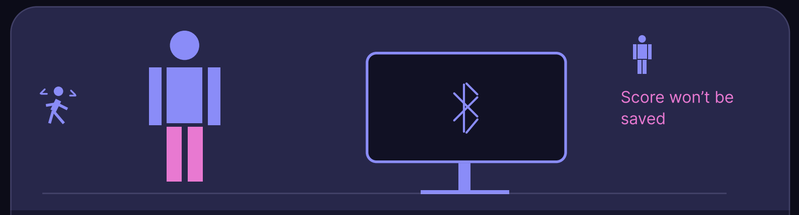
01 Take a break

After studying, a student opens AYATLA and clears enough space to move.



02 Calibrate

They enter the body guide and hold still while the camera detects their body and hands.



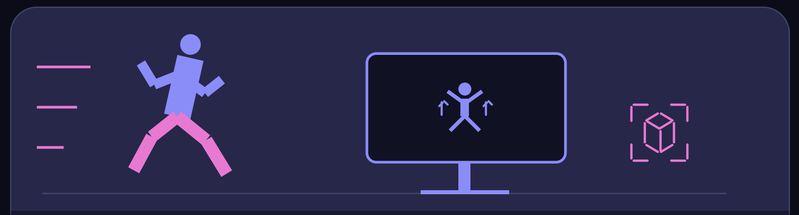
03 Choose a mode

They lean left to sync a Bluetooth profile or right to continue as a guest.



04 Start

They use a numbered finger gesture to focus Start, then make an OK sign to confirm.



05 Run

They lean, jump, and crouch while the runner mirrors their movements.



06 Finish

They hold a flat palm to pause, choose an option with their fingers, and use the OK sign to confirm.

“A complete game session — without touching the laptop once.”



RESEARCH, LEARN, IMPROVE

Survey & Critique

A proposed discovery study — questions, not claimed findings.

AYATLA · HUMAN–COMPUTER INTERACTION

PROPOSED DISCOVERY RESEARCH

AYATLA discovery survey

A study plan to validate the interaction — not collected research.

Target · 12–35

Proposed survey · No results

WHO

Occasional gamers with a webcam-equipped computer.

PURPOSE

Test comfort, context, and expectations for gesture control.

METHODOLOGY

Balanced Likert scales, neutral wording, and one topic per question.

Sample answer states on the next pages are illustrative only. They are not participant responses or research findings.

WHY THESE QUESTIONS WORK

01–03	Profile users and prior experience.
04–05	Test comfort and physical context.
06–08	Guide gestures, menu selection, and feedback.
09	Set session pacing.
10	Surface unanticipated concerns.

Camera privacy matters

Explain what is processed, what is stored, and when the camera is active.

AYATLA RESEARCH

Hands-free gaming survey

10 QUESTIONS · 4 MIN

PART 1 OF 3 · PROPOSED SURVEY

- Violet sample selected states are illustrative only — not research results.

01 What is your age group?

○ Under 12 ○ 12–17 ● 18–24 ○ 25–35 ○ Over 35

02 How often do you play video games?

○ Never ○ Rarely ○ Monthly ● Weekly ○ Daily

03 Have you used a motion-controlled system?

○ Yes ● No ○ Not sure

AYATLA RESEARCH

Hands-free gaming survey

10 QUESTIONS · 4 MIN

PART 2 OF 3 · PROPOSED SURVEY

- Violet sample selected states are illustrative only — not research results.

04 How comfortable are you with body-only controls?

○ Very uncomfortable ○ Uncomfortable ● Neutral ○ Comfortable ○ Very comfortable

05 How much space is available in front of your computer?

○ Less than 1 m ● 1–2 m ○ More than 2 m ○ Not sure

06 Which gesture feels most natural for pausing?

● Flat palm ○ Closed fist ○ Waving ○ Clapping ○ Other

07 Which method feels clearest for selecting a menu option?

● Finger count ○ Pointing ○ Leaning ○ Hand dwell ○ Other

AYATLA RESEARCH

Hands-free gaming survey

10 QUESTIONS · 4 MIN

PART 3 OF 3 · PROPOSED SURVEY

- Violet sample selected states are illustrative only — not research results.

08 How important is a live camera preview during gameplay?

- ○ Not important
- ○ Slightly important
- ● Moderately important
- ○ Very important
- ○ Extremely important

09 How long would a typical play session last?

- ○ Under 5 min
- ● 5–15 min
- ○ 15–30 min
- ○ Over 30 min

10 What concerns do you have about a camera-based game?

OPEN RESPONSE

Share any concerns in your own words...

CRITIQUE & POTENTIAL ISSUES

What the survey may miss

01 Hypothetical bias

Users may not predict comfort accurately before trying gesture control.

POSSIBLE FIX

Follow up with a prototype usability test.

02 Leading answer choices

Gesture questions may steer users toward existing interaction ideas.

POSSIBLE FIX

Ask openly first, then show options.

03 Inaccurate space estimates

Self-reported room dimensions may be unreliable.

POSSIBLE FIX

Add visual examples for each distance.

04 Sampling bias

An online survey may over-represent young, tech-savvy users.

POSSIBLE FIX

Recruit through several channels, including in person.

05 Accessibility gap

The survey does not fully identify users with limited mobility.

POSSIBLE FIX

Add an optional physical-access question.

06 Small sample risk

A short project timeline may limit confidence in findings.

POSSIBLE FIX

Set a minimum target of 30 responses.

AYATLA

Move naturally. Play completely hands-free.

HCI PROJECT · 2026

BODY / HANDS / OBJECTS